

BUSINESS SYSTEMS ANALYST INTERNSHIP

Practical Assignment: System Analysis & Troubleshooting

Company: The Flying Panda

1. Workflow Analysis:

Assumptions: The system uses an online enquiry form, a database, a team dashboard, a booking module and an email notification service.

Stage	Input	Expected Output
Enquiry submission	Customer name, email and message	Enquiry submitted.
Database storage	Submitted enquiry	One saved record with a unique ID
Team assignment	Saved enquiry	Enquiry assigned to a team member
Consultation booking	Date, time and enquiry ID	Appointment saved
Confirmation	Successfully saved booking	Confirmation sent to customer

System interactions: The customer submits the form, and the application stores the enquiry in the database. The enquiry is assigned to a team member. When the customer books a consultation, the booking module saves the appointment and the notification service sends confirmation.

Each step should be completed successfully before the system confirms the corresponding enquiry or booking.

2. Root Cause Analysis:

Issue A: Duplicate Enquiries

Cause 1: Multiple form submissions. A customer may click the submit button more than once, creating duplicate records.

Investigation: Review application logs and database records for repeated requests. Test by submitting the same form twice quickly.

Cause 2: Automatic request retries. A request may be retried after a timeout, creating another record.

Investigation: Check request and retry logs to see whether repeated requests generate separate enquiry IDs. Verify whether duplicate-prevention controls exist.

Issue B: Confirmation Received but Enquiry Missing

Cause 1: Confirmation sent before saving is completed. The application may show success before verifying that the enquiry is saved.

Investigation: Compare application logs, database records and notification timestamps. Check the order of operations.

Cause 2: Database save failure. A connection error or database constraint may prevent the enquiry from being saved, while the application still reports success.

Investigation: Review server and database error logs and verify whether the enquiry record exists.

Issue C: Booking Confirmation Received but Appointment Missing

Cause 1: Booking save failure. The system may send confirmation even though the appointment was not saved.

Investigation: Compare booking records with notification logs. Simulate a save failure in a test environment and check whether confirmation is sent.

Cause 2: Dashboard synchronization or retrieval problem. The appointment may be saved but not displayed because of incorrect filters, outdated data or a retrieval error.

Investigation: Check the database record, appointment status, assigned team member and dashboard logs. Refresh the dashboard and verify whether the appointment appears.

These are possible causes; investigation is required to confirm the actual cause.

3. Test Cases:

Test case and steps	Expected result
TC01 – Normal enquiry: Submit a valid enquiry once.	Exactly one record is saved and visible to the team.
TC02 – Duplicate submission: Submit the same form twice quickly.	No unintended duplicate record is created.
TC03 – Database failure: Simulate a save failure and submit an enquiry.	No success confirmation is shown; an appropriate error or retry option appears.
TC04 – Successful booking: Book an available consultation slot.	One appointment is saved, visible to the team and confirmed to the customer.
TC05 – Appointment visibility: Save a booking and refresh the team dashboard.	The appointment appears with the correct date, time and status.

4. Bug Report:

Bug Title: Success confirmation shown when an enquiry is not saved.

Precondition: The enquiry form is available, and a database save failure can be simulated in a test environment.

Steps to Reproduce:

1. Open the enquiry form and enter valid details.
2. Simulate a database save failure.
3. Submit the form.
4. Check the success message and search for the enquiry in the database or dashboard.

Expected Result: Success should be displayed only after the enquiry is saved. If saving fails, the system should show an error and allow the customer to retry.

Actual Result: In the reported scenario, the customer receives confirmation, but the enquiry is missing.

Severity: High. Missing enquiries can prevent customer follow-up and reduce trust. The behaviour should be reproduced to confirm the bug.

5. Suggested Improvements:

1. Reliable submission and duplicate prevention: Save the enquiry before sending confirmation. Use a unique request identifier or suitable database constraints to prevent duplicate records. Display a clear error if saving fails.

2. Reliable booking and monitoring: Confirm bookings only after successful saving. Monitor whether appointments appear on the team dashboard and log failures in booking, notification and synchronization processes so the team can investigate them promptly.

Conclusion

The system should save each enquiry and booking correctly, display records to the team and send confirmations only after successful processing. Proper error handling, duplicate prevention and regular testing can improve system reliability and customer experience.